

Operators, precedence and expressions ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Classify arithmetic, relational, logical, assignment, increment, conditional, bitwise operators
- Apply precedence: () then ++/-- then * / % then + -
- Trace small programs with x++ vs ++x

Class	Operators	Result
Arithmetic	+ - * / %	% remainder, integers only
Relational	> < >= <= == !=	1 (true) or 0 (false)
Logical	&& !	compound conditions
Assignment	= += -= *= /= %=	x += 2 means x = x+2
Increment	++ --	pre or post
Conditional	? :	x = (a>b)? a : b;
Bitwise	& ^ ~ << >>	bit level (rare in Paper II but listed in books)

LEARN IT

Precedence (high → low): () ++ -- * / % + - relational logical assignment. Same-level * / % go left to right.

Integer division: 5/2 is 2, not 2.5. Use 5/2.0 or a float variable.

WORKED EXAMPLE / BOARD STYLE

If x = 1:

printf("x=%d\n", x); → x=1

printf("x=%d\n", x + x); → x=2

printf("x=%d\n", x); → x=1 (x was never assigned a new value)

Post-increment example: k++ is post; ++k is pre. MCQ: k++ is post increment.

```
int a = 5, b = 2, c;
c = a / b;           /* 2 */
c = a % b;           /* 1 */
c = (a > b) && (b > 0); /* 1 */
c = (a > b) ? a : b;   /* 5 */
```

HOW BIEK ASKS THIS

Long 2026 Q4: types of operators in C. Make a table with one example each. Trace questions: never change a variable unless you see =, ++ or --.

CHECK YOURSELF

1. Value of 17 % 5?
2. Is k++ pre or post increment?

Answers: 1. 2 · 2. Post

MCQs

1. An example of post increment is:
 - a. k++
 - b. ++k
 - c. k+=
 - d. k=+
2. Highest precedence among + - / % is:
 - a. +
 - b. -
 - c. /
 - d. % — actually / and % are equal and higher than + -
3. == is a:
 - a. assignment
 - b. relational operator
 - c. loop
 - d. header

Key: 1=a, 2=c, 3=b

Short questions (3 marks each)

1. Define increment and decrement with examples.
2. If x=1, output of three printf lines using x, x+x, x.
3. Write C expressions for $(x^2 - y^2)(x+y)/(a+b)$.

Model short answers

1. ++k pre-increment, k++ post. -- same for decrement. int k=5; printf("%d", k++); prints 5 then k becomes 6.
2. 1 then 2 then 1.
3. $((x*x - y*y)*(x+y))/(a+b)$

Long questions (board Section C)

1. What are different types of operators used in a C program? Explain with examples and precedence.